Colégio
Agostiniano Mendel

ATIVIDADE DE APOIO – ENSINO MÉDIO

Componente Curricular: Física-Dinâmica **Data:** Fevereiro/2026
Professor (a): Rosana-Pádua **Série:** 2º
Assunto (s): Força de Atrito II- Exercícios para Estudar

---

1. • $Fat_{em} = \mu e \cdot N$ | $Fat_{em} = 0{,}3 \cdot 40$ | $Fat_{em} = 12N$

• $Fat_d = \mu d \cdot N$ | $Fat_d = 0{,}25 \cdot 40$ | $Fat_d = 10N$

* $P = 40N \Rightarrow m = 4kg$

• $F = 10N \Rightarrow F < Fat_{em} \Rightarrow$ bloco em repouso $(F_R = 0)$ $\therefore a = 0$
$Fat_e = F \Rightarrow$ $Fat_e = 10N$

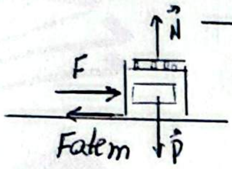
• $F = 12N \Rightarrow F = Fat_{em} \Rightarrow$ bloco em repouso c/ iminência de mov.
$a = 0$ e $F = Fat_{emax} \Rightarrow$ $Fat_{emax} = 12N$

• $F = 30N \Rightarrow F > Fat_{em} \Rightarrow$ bloco em mov $\Rightarrow Fat_d = 10N$

∘ $F_R = m \cdot a \Rightarrow F - Fat_d = m \cdot a \Rightarrow 30 - 10 = 4 \cdot a \Rightarrow 20 = 4a$
$a = 5 m/s^2$

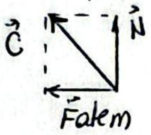
---

2. $m = 40kg$
$\mu e = 0{,}75$



a) * Iminência de escorregar $\Rightarrow$ repouso $\Rightarrow$ $F_R = 0$
$F = Fat_{em} \Rightarrow F = \mu e \cdot N \Rightarrow F = 0{,}75 \cdot 400 \Rightarrow$ $F = 300N$

b) Força de Contato $\Rightarrow$ força recebe do solo

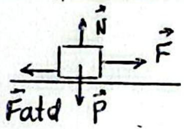


$C^2 = N^2 + Fat^2 \Rightarrow C^2 = (400)^2 + (300)^2$
$C^2 = 250000 \Rightarrow C = \sqrt{25 \cdot 10^4}$
$C = 500N$

---

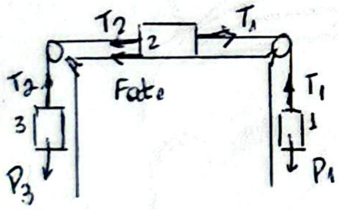
3. $m = 1500kg$
$F = 750N$

* MU $(F_R = 0)$



$F = Fat_d$
$F = \mu d \cdot N \Rightarrow 750 = \mu d \cdot 15000 \Rightarrow \mu d = 0{,}05$

4.



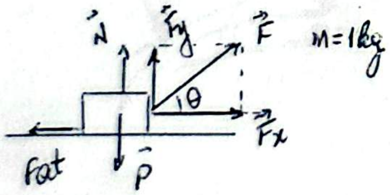
$P_1 = 60N$
$P_2 = 100N$
$P_3 = 40N$

Sistema em repouso $\Rightarrow F_R = 0$ (Fat_e para a esquerda)

$1 \Rightarrow P_1 = T_1 \Rightarrow T_1 = 60N$
$3 \Rightarrow P_3 = T_2 \Rightarrow T_2 = 40N$

$\therefore$ $2 \Rightarrow T_1 = T_2 + Fat_e$
$60 = 40 + Fat_e \Rightarrow$ $Fat_e = 20N$

5.



$m = 1kg$

• $N + F_y = P$
$N + 6 = 10$
$N = 4N$

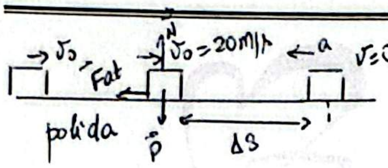
• $Fat = \mu \cdot N$
$Fat = 0{,}25 \cdot 4$
$Fat = 1N$

• $F_x - Fat = m \cdot a$
$8 - 1 = 1 \cdot a$
$a = 7 m/s^2$

• $F_x = F \cdot \cos\theta \Rightarrow F_x = 10 \cdot 0{,}8 = 8N$

• $F_y = F \cdot \sin\theta \Rightarrow F_y = 10 \cdot 0{,}6 = 6N$

6.



- $F_{at} = m.a$

$\mu \cancel{m} g = \cancel{m} a$

$a = \mu . g \Rightarrow a = 0,4.10$

$a = 4 m/s^2$

- $v^2 = v_0^2 + 2a\,\Delta S$

$0 = 20^2 + 2.(-4).\Delta S$

$8\,\Delta S = 400$

$\Delta S = 50m \quad \therefore \; d = |\Delta S| \Rightarrow \underline{d = 50m}$

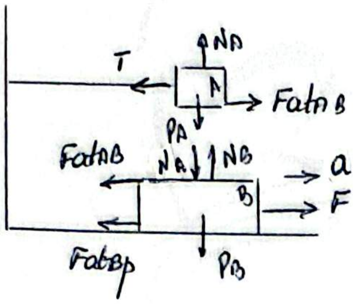
7.

$\mu_{AB} = 0,2$

$\mu_{Bp} = 0,3$

$m_A = 5kg$

$m_B = 15kg$



- $N_A = P_A \Rightarrow N_A = 50N$
- $N_B = N_A + P_B$

$N_B = P_A + P_B$

$N_B = 50 + 150 \Rightarrow N_B = 200N$

- $Fat_{AB} = \mu_{AB}\, N_A$

$Fat_{AB} = 0,2.50 \Rightarrow Fat_{AB} = 10N$

- $Fat_{Bp} = \mu_{Bp}.N_B$

$Fat_{Bp} = 0,3.200 \Rightarrow Fat_{Bp} = 60N$

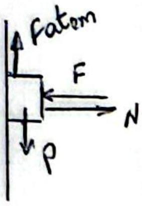
a) $A \Rightarrow$ repouso ($F_R = 0$)

$T = Fat_{AB} \Rightarrow \underline{T = 10N}$

b) $F_R = m.a \Rightarrow F - Fat_{AB} - Fat_{Bp} = m_B.a \quad (a = 1 m/s^2)$

$F - 10 - 60 = 15.1 \Rightarrow \boxed{F = 85N}$

8.



Iminência de movimento ⇒ repouso ⇒ $F_R = 0$

* $\underline{N = F}$ (horizontal; repouso e $F_R' = 0$)

* Vertical ⇒ $Fat_{em} = P \Rightarrow \mu e.N = P \Rightarrow \mu e.F = P$

$0,3.F = 60 \Rightarrow \underline{F = 200N}$